
Neural Morphogen Operators: Learning Continuous Reaction–Diffusion Dynamics from Spatial Transcriptomics

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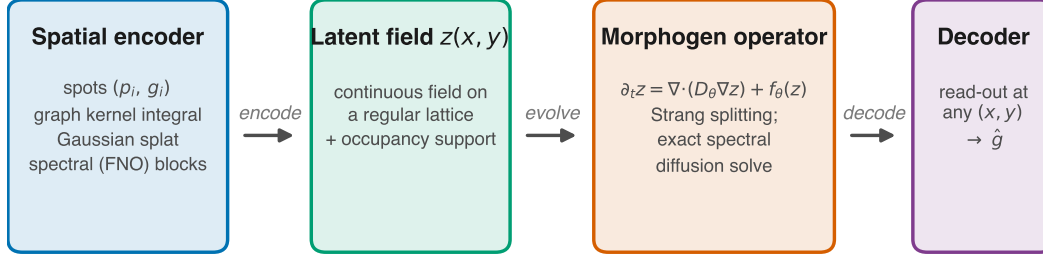
Abstract

Spatial transcriptomics is almost universally modelled as a graph over measured spots, an inductive bias that ties the learned function to one sampling lattice and admits no notion of expression between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. We introduce **Neural Morphogen Operators** (NMO), which encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting removes the CFL constraint and renders $\mathbf{D}_\theta$ as a physical length scale (374 μm median). On 9 sections from 6 public datasets spanning four technologies, NMO improves masked-region reconstruction over graph and Gaussian-process baselines ($r = 0.247 \pm 0.005$ vs. 0.234 ± 0.001). Two controls locate the effect: an exactly parameter-matched ablation shows the 0.43% of weights forming the operator account for -0.023 Pearson r , while a non-spatial baseline shows graph message passing adds at most 0.005 over distance-weighted interpolation against 0.017 for the operator. Three further tests are negative or null: the fitted operator is dissipative with no Turing instability at any finite wavenumber, its fields are the most over-smoothed of any model tested, a pre-specified split-half perturbation test does not reach significance, and it transfers *worse* than the baseline it beats in-domain (2.8–2.0 $\times$). The continuous-operator view buys in-domain accuracy by specialising to a single tissue; its portability and mechanistic readings remain unsupported.

1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field: morphogens are produced, diffuse, decay and interact, and cells read the resulting landscape [Turing, 1952, Wolpert, 1969,



trained by masked spatial reconstruction; held-out regions are never seen by the encoder

Figure 1: **Neural Morphogen Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction–diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure must pass through the field and therefore through the operator.

33 Gierer and Meinhardt, 1972]. Reaction–diffusion models of this form account for phenomena from
 34 digit patterning [Sheth et al., 2012] to Nodal/Lefty signalling [Müller et al., 2012], with measured
 35 gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue
 36 organisation is generated by a continuous process, a model of that process—not of the lattice it was
 37 sampled on—is the natural object to fit.

38 We test that proposition directly. **Neural Morphogen Operators** (NMO) encode irregularly sam-
 39 pled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion
 40 PDE, and decode at arbitrary coordinates (Fig. 1). We ask where this helps and where it does not. We
 41 are *not* asking whether the fitted operator recovers biochemistry; a single snapshot cannot support
 42 that, for reasons made precise in Section 7.

43 **Contributions.** (i) A PDE-constrained neural operator for spatial omics coupling a graph kernel-
 44 integral encoder, a spectral reaction–diffusion operator and a coordinate-free decoder (Section 3).
 45 (ii) An unconditionally stable, interpretable integrator: diffusion applied exactly in the Fourier do-
 46 main under Strang splitting, removing the CFL constraint and rendering $\mathbf{D}_\theta$ as a diffusion length in
 47 microns. (iii) A correct formulation of the physics loss—because the integrator is exact the usual
 48 PDE residual is vacuous, and the content-bearing constraint is a *steady-state* residual. (iv) Two con-
 49 trols that locate the effect: an exactly parameter-matched $T=0$ ablation, and a non-spatial baseline
 50 showing how little graph message passing contributes. (v) Four negative results reported as results—
 51 no Turing instability, worst-in-class spatial autocorrelation, a failed split-half perturbation test, and
 52 worse transfer than the baseline we beat in-domain.

53 2 Related Work

54 **Neural operators** learn maps between function spaces, giving approximate independence from the
 55 discretisation [Kovachki et al., 2023]: Fourier neural operators parameterise the kernel spectrally [Li
 56 et al., 2021], graph neural operators integrate a kernel over neighbourhoods on irregular meshes [Li
 57 et al., 2020], DeepONet factorises into branch and trunk networks [Lu et al., 2021], and message-
 58 passing solvers learn the numerical stencil itself [Brandstetter et al., 2022]. We use the graph form
 59 for irregular sampling and the spectral form for the dynamics, because the latter makes the diffusion
 60 solve exact; the integrator follows classical exponential time-differencing and operator splitting [Cox
 61 and Matthews, 2002, Strang, 1968] rather than soft PDE-residual penalties [Raissi et al., 2019] or
 62 neural ODEs [Chen et al., 2018].

63 **Machine learning for spatial transcriptomics** is dominated by graph models built for domain
 64 identification and denoising rather than prediction at unmeasured coordinates—SpaGCN [Hu et al.,
 65 2021], STAGATE [Dong and Zhang, 2022], SpatialDE/SPARK [Svensson et al., 2018, Sun et al.,

2020], DestVI [Lopez et al., 2022]—which is why we adapt them (Section 4); Spateo [Qiu et al., 2024] is closest in spirit but targets 3-D reconstruction rather than a transferable operator. **Pattern formation** supplies our vocabulary [Turing, 1952, Kondo and Miura, 2010, Green and Sharpe, 2015], while RNA velocity [Bergen et al., 2020] and perturbation models [Lotfollahi et al., 2019, Roohani et al., 2024] model dynamics or interventions without an explicit spatial operator—the gap we address.

3 Method

A section provides N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalised to $[-1, 1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \rightarrow \mathbb{R}^C$ and factor the model as encode–evolve–decode.

Spatial encoder. Irregular sampling is handled by kernel-integral layers on a k -NN graph [Li et al., 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement to a per-channel gain; depending on geometry rather than node identity is what lets an encoder fitted to one lattice be evaluated on another. Features are splatted onto a regular $H \times W$ lattice (Nadaraya–Watson, learnable bandwidth), yielding a field and an *occupancy* map that distinguishes “tissue with no signal” from “no measurement here” — the distinction the masked-region task hinges on. Spectral blocks [Li et al., 2021] then refine the field.

The morphogen operator. The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z}), \quad (1)$$

with a per-channel symmetric positive-definite tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so definiteness is structural rather than penalised) and a pointwise reaction network f_θ implemented as 1×1 convolutions with a tanh-bounded output.

Equation (1) is stiff—the diffusion operator has eigenvalues scaling as $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t = O(h^2/\|\mathbf{D}\|)$ —but for constant-coefficient anisotropic diffusion it is diagonal in the Fourier basis, so evolution over τ has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \tau) \mathcal{F}. \quad (2)$$

Combined with a midpoint reaction update under Strang splitting [Strang, 1968] this is second-order accurate, with two consequences. Since $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (2) lies in $(0, 1]$ for every Δt : the diffusive channel is **unconditionally stable** whatever the optimiser does to θ , and empirically rollouts remain bounded for 10^3 steps, two orders of magnitude beyond the training horizon. And $\mathbf{D}_c$ is interpretable—its eigenvalues are squared diffusion lengths per unit pseudo-time, convertible to microns through the stored coordinate scale.

Decoder. Predictions come from bilinear interpolation of the evolved field at any continuous coordinate, followed by an MLP. *The decoder receives no positional input:* given coordinates it could memorise a coordinate-to-expression map, rendering the dynamics ornamental and every ablation uninterpretable.

Objectives, and what a physics loss can mean here. The objective combines reconstruction (MSE plus a per-gene Pearson term, since the spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and three physics terms. Because the integrator solves (1) essentially exactly, the residual along the model’s own trajectory is zero up to splitting error, so imposing it would penalise the integrator rather than the model. The constraint with content is that the *measured configuration should be a near-stationary state* of the learned operator:

$$\mathcal{L}_{\text{PDE}} = \|\nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\|^2, \quad (3)$$

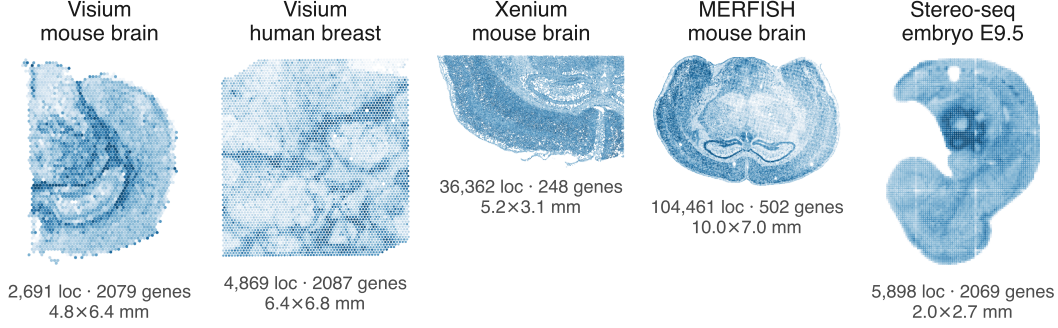


Figure 2: **Processed spatial sections**, coloured by total counts per location. The five sections differ in assay, sampling density (2.7k Visium spots to 104k MERFISH cells), gene panel (248–2087) and physical extent (2–10 mm), which is precisely what the transfer experiments vary. Full inventory in Appendix C.

the correct formalisation of the quasi-steady-state assumption for adult tissue, constraining $\mathbf{D}_\theta$ and f_θ jointly. We add mass, Jacobian-norm and splitting-consistency terms; the ablations show all three are inert at our weights.

Interpretability: dispersion analysis. Linearising (1) about a homogeneous $\mathbf{z}^*$ gives $\partial_t \delta \mathbf{z}_k = (\mathbf{J} - \mathbf{k}^\top \mathbf{D} \mathbf{k}) \delta \mathbf{z}_k$ with $\mathbf{J} = \partial f_\theta / \partial \mathbf{z}|_{\mathbf{z}^*}$, so $\max \text{Respec}(\mathbf{J} - |\mathbf{k}|^2 \mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the *learned* operator (maximised over orientations of $\mathbf{k}$, since $\mathbf{D}$ is anisotropic). Positive growth at finite $|\mathbf{k}|$ with decay at $\mathbf{k} = 0$ is the classical Turing signature: a falsifiable test rather than a post-hoc visualisation.

4 Experimental setup

Data. 6 public datasets span four spatial technologies plus one non-spatial perturbation screen (9 sections, 469,038 locations; Fig. 2). Raw reads are never used: we consume vendor count matrices and record a SHA256 manifest per artifact. One QC policy is applied throughout, then library-size normalisation, log transform, and HVG selection that *force-include*s every detected morphogen-pathway gene, so the interpretability analysis is not evaluated on a set selection already biased. Coordinates are normalised **isotropically**: anisotropic rescaling would distort the Laplacian and make a learned diffusion coefficient meaningless.

Task and splits. Masked spatial reconstruction: contiguous tissue blocks are hidden from the encoder and every model predicts expression at those coordinates. Random spot-level splits leak, since a held-out spot is trivially predicted from its correlated neighbours, so we hold out whole contiguous blocks [Roberts et al., 2017] and compute standardisation statistics on the training split only.

Baselines. A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani et al., 2017], and an exact GP [Rasmussen and Williams, 2006]. The graph methods were not designed to predict at unmeasured coordinates, so each receives the same inverse-distance read-out head—the most favourable choice available. We mark them *-style* rather than claiming they reproduce published numbers. All models share one training loop, masks, optimiser and evaluation code.

Metrics. Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al., 2004] on rasterised maps, and absolute error in Moran’s I [Moran, 1950]. The last is included

Table 1: Masked spatial reconstruction on visium_mouse_brain. Contiguous tissue blocks are hidden from every model; metrics are computed on held-out locations only. Mean $\pm$ s.d. over 3 seeds. All models share the identical training loop, masks and evaluation code. $|\Delta I|$ is the mean absolute error in Moran’s I between predicted and measured expression maps: it penalises over-smoothing that correlation alone does not detect.

model	params	Pearson $r \uparrow$	Spearman $\rho \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$	$ \Delta I \downarrow$
SpaGCN-style	0.59M	0.234 ± 0.001	0.184 ± 0.001	0.254 ± 0.001	0.393 ± 0.001	0.738 ± 0.002
STAGATE-style	0.69M	0.234 ± 0.002	0.185 ± 0.002	0.254 ± 0.001	0.394 ± 0.002	0.737 ± 0.003
Graph Transformer	0.83M	0.230 ± 0.000	0.182 ± 0.001	0.255 ± 0.001	0.390 ± 0.000	0.723 ± 0.004
Autoencoder (non-spatial)	0.58M	0.230 ± 0.001	0.180 ± 0.002	0.255 ± 0.000	0.387 ± 0.002	0.727 ± 0.001
GNN (GCN)	0.61M	0.228 ± 0.001	0.181 ± 0.001	0.253 ± 0.001	0.386 ± 0.002	0.755 ± 0.002
Gaussian Process	0.00M	0.149 ± 0.008	0.108 ± 0.006	0.293 ± 0.004	0.297 ± 0.012	0.718 ± 0.003
NMO (ours)	2.20M	0.247 ± 0.005	0.203 ± 0.003	0.252 ± 0.004	0.404 ± 0.006	0.762 ± 0.004

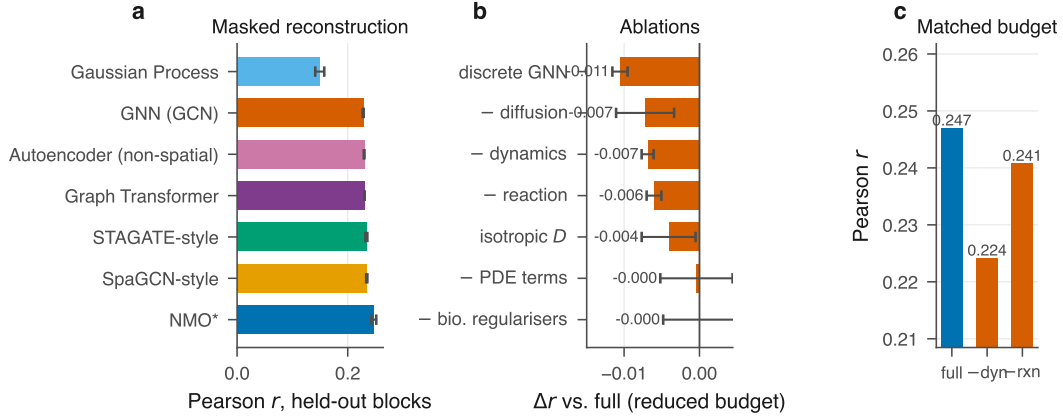


Figure 3: **Where the effect comes from.** (a) Masked-region reconstruction on the adult mouse-brain Visium section; NMO leads, but the graph baselines cluster within 0.007 of the *non-spatial* autoencoder. (b) Ablation deltas at the reduced budget: structural terms matter, regularisers do not. (c) The two decisive controls at the benchmark budget—disabling the operator ($T=0$) costs more than NMO’s entire margin over the best baseline and drops it below every model in (a).

136 because correlation and RMSE both reward regressing towards a smooth mean. Mean $\pm$ s.d. over 3
137 seeds.

138 5 Results

139 5.1 Masked spatial reconstruction

140 NMO reaches Pearson r of 0.247 ± 0.005 on held-out blocks against 0.234 ± 0.001 for the strongest
141 baseline (spagcn), a gain of 0.013 (Table 1, Fig. 3a). The margin is modest and the graph baselines
142 cluster tightly, so the ablations are the stronger evidence. The Gaussian process underperforms
143 them—a property of our baseline, not of GP regression, since one RBF length scale is shared across
144 all genes.

145 **Graph message passing barely beats plain interpolation.** The most informative row is the one
146 included as a floor. The non-spatial autoencoder sees no coordinates at all and reaches 0.230 purely
147 through the inverse-distance read-out head shared by every graph baseline, so the gap between it
148 and a graph model isolates what *message passing* contributes. That contribution is close to nothing:
149 across 4 graph models the margin spans -0.002 to 0.005 Pearson r , and 1 is worse than using no
150 spatial information at all, while NMO gains 0.017 — roughly $3.6\times$ the best graph model’s mar-

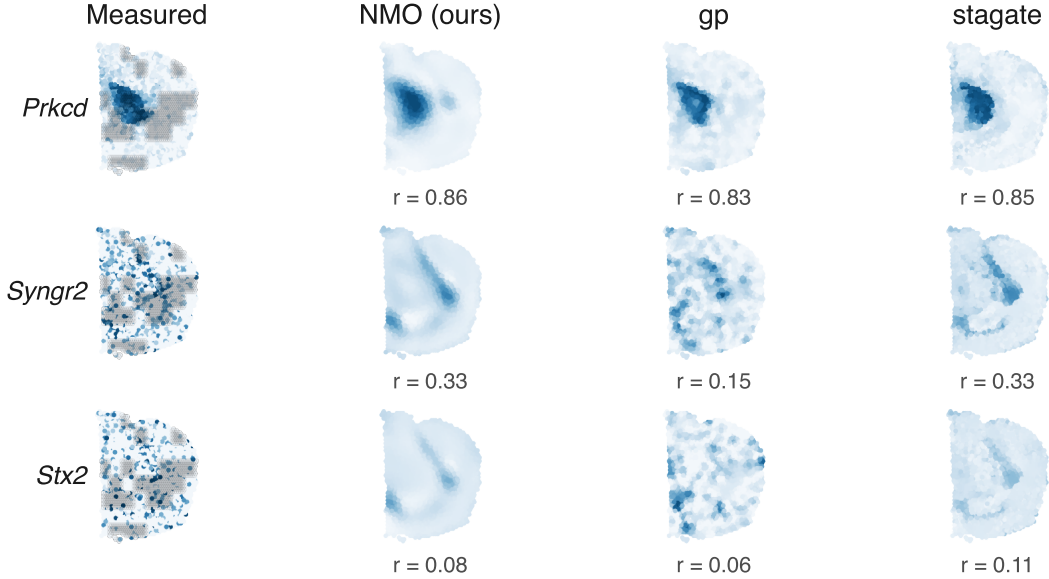


Figure 4: **Masked reconstruction across the performance range.** Measured expression (left; held-out spots outlined) against NMO and two baselines, for genes at the best, median and lower-quartile of per-gene Pearson r . Predicted fields are visibly smoother than the measurement—the bias that Moran’s I quantifies in Table 1.

gin. On this task the standard graph treatment extracts remarkably little beyond distance-weighted interpolation; we would encourage the control to be reported routinely.

One metric disagrees, and it is the one added as a guard. NMO is best on Pearson, Spearman, RMSE and SSIM (0.404 ± 0.006 against 0.394), but *worst* on Moran’s I error: 0.762 ± 0.004 against 0.718, a gap of +0.044. SSIM measures local contrast on a rasterised map while Moran’s I measures global autocorrelation on the point set, and NMO’s local structure is good while its fields are far too autocorrelated overall ($I = 0.936 \pm 0.004$ against a measured 0.174; Fig. 4). The cause is the dissipative operator (Fig. 5c), which attenuates high spatial frequencies—helpful inside a masked block, harmful to texture.

5.2 Learned fields and dynamics

The fitted operator has a median diffusion length of $374 \mu\text{m}$ (10–90th pct. 295–478), the same order as measured morphogen gradients [Kicheva et al., 2007, Yu et al., 2009] — a consistency check, not a measurement of physical transport.

The learned operator is not Turing-unstable. The dispersion relation (Fig. 5c) decreases monotonically in $|\mathbf{k}|$: growth is weakly positive only for the uniform mode and strictly negative at every finite wavenumber, so no band of spatial modes is spontaneously amplified and no characteristic wavelength exists. Fitted operators were Turing-unstable in 0% of runs. From a stationary snapshot the model learns a *dissipative* operator whose fixed point is the observed tissue—exactly what (3) asks for. The absence of a Turing band reflects the data available, not evidence against reaction–diffusion accounts of the biology; detecting genuine pattern-forming instability would require observing tissue away from steady state.

5.3 Transfer across tissue, species and resolution

This is the clearest negative result in the paper. Trained on adult mouse brain and applied unchanged to human breast carcinoma—a different species, organ and spatial architecture—NMO

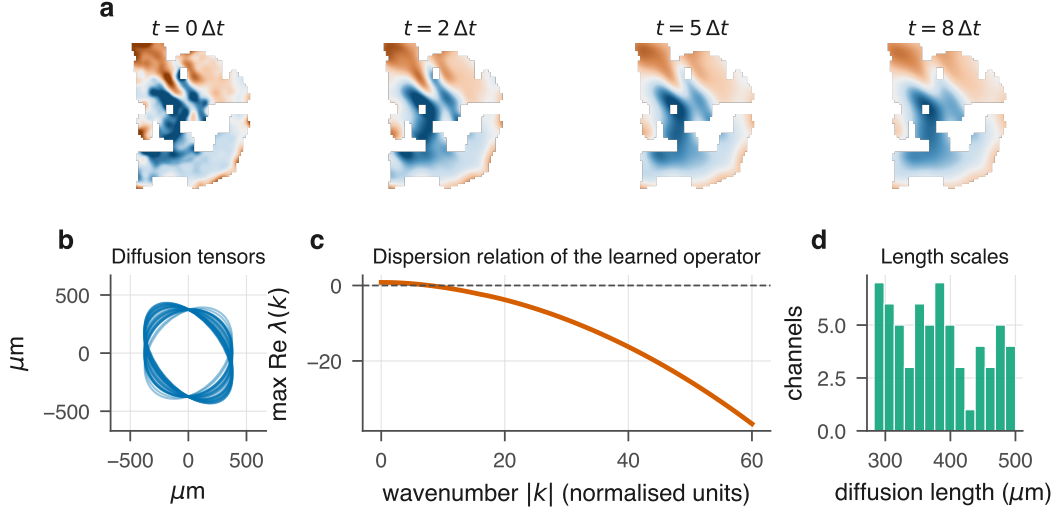


Figure 5: **What the fitted operator looks like.** (a) A latent channel relaxing under the learned dynamics. (b) Per-channel diffusion tensors as length-scale ellipses, showing the anisotropy is used. (c) Dispersion relation: growth is negative at every finite wavenumber, so the operator is dissipative and supports no Turing band. (d) Diffusion lengths across channels.

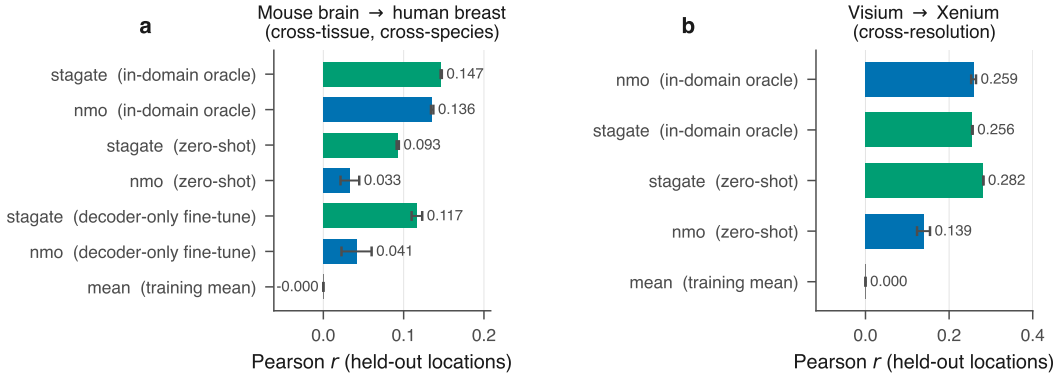


Figure 6: **Transfer is where the operator fails.** Zero-shot performance against the in-domain oracle and the training-mean floor, for (a) mouse brain → human breast carcinoma and (b) Visium → single-cell Xenium. The STAGATE-style baseline that NMO beats in-domain transfers substantially better in both settings.

attains 0.033 ± 0.012 zero-shot against a training-mean floor of -0.000 ± 0.000 and an in-domain oracle of 0.136 ± 0.001 . It is above the floor, so something transfers. But stagate-style reaches 0.093 ± 0.001 in the same setting, $2.8\times$ better than the model that beat it in-domain (Fig. 6); refitting only the decoder does not close the gap (0.041 ± 0.019). Cross-resolution repeats the pattern: an operator fitted to $55 \mu\text{m}$ Visium spots and evaluated on single-cell Xenium sampling of the *same organ* reaches 0.139 ± 0.015 against 0.282 ± 0.000 , again $2.0\times$ worse (Appendix F). All Xenium panel genes are present in the raw Visium matrix, so we rebuild the source with the panel force-included (248 shared genes); the gap is not an artefact of thin gene overlap.

We expected the opposite, since an operator defined on the field rather than a lattice should tolerate changes of geometry. The most plausible explanation—a hypothesis, not a demonstration—is that the advantage and the failure share a cause: NMO wins in-domain by fitting an operator whose fixed point *is* this tissue, and relaxing towards that attractor is wrong when the target has a different architecture, whereas local message passing carries less about the source. Discretisation-invariance claims for neural operators should not be assumed to extend to biological transfer without test.

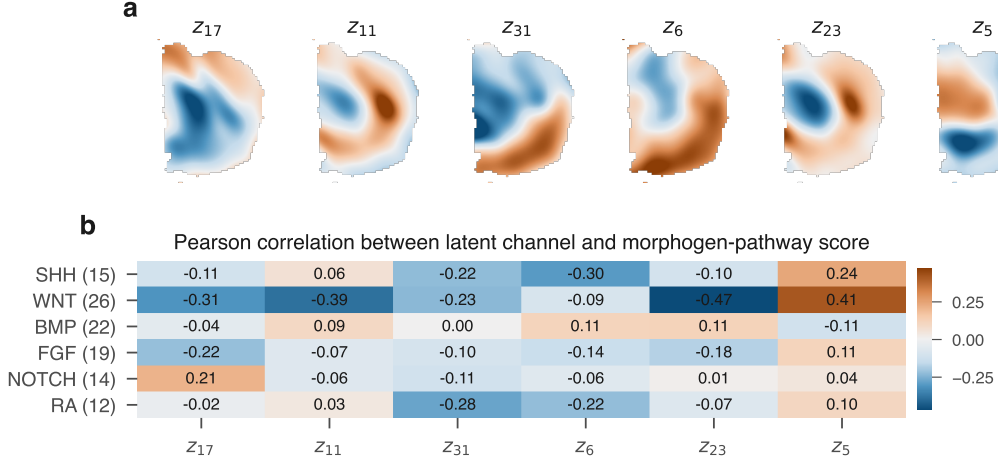


Figure 7: **Latent fields align with anatomy, not with pathways.** (a) Latent channels of the relaxed field recover laminar and nuclear structure. (b) Correlation between each channel and canonical morphogen-pathway scores. These correlations are real, but do not survive the causal test.

189 5.4 Developmental forecasting

190 The Stereo-seq series is the only dataset here with a real time axis. Integrating the encoded E9.5
 191 field forward improves the field-level match to E10.5 from 0.082 at $T=0$ to 0.095 at $T=16$, against
 192 0.064 for predicting no change: the learned operator carries genuine temporal information, with a
 193 sensible optimum horizon. Consecutive MOSTA stages are *different embryos*, so this is a field-level
 194 comparison after coarse registration (Appendix H).

195 5.5 Counterfactual perturbation

196 We test the pathway interpretation with a deliberately falsifiable split-half design: inject the latent
 197 direction that maximally raises one random half of a morphogen pathway, integrate the operator, and
 198 ask whether the *held-out* half responds preferentially, against a permutation null over matched gene
 199 sets.

200 **It fails.** Across 4 pathways and 2 seeds, 0 reach significance (smallest $p = 0.47$) and the mean
 201 held-out rank is 59% against a 50% chance level—held-out pathway members respond, if anything,
 202 slightly *less* than random genes. This rules out the reading that Fig. 7b would otherwise invite: those
 203 correlations are consistent with the field encoding coarse anatomy that pathway scores also track, so
 204 we do not claim NMO recovers signalling programmes. The experiment does establish a scale —
 205 the response decays to half its peak within $577 \mu\text{m}$, matching the diffusion lengths read off $\mathbf{D}_\theta$. It
 206 is the *specificity*, not the range, that is rejected. A second pre-specified probe against Perturb-seq
 207 [Norman et al., 2019] was underpowered ($n = 7$) and is inconclusive (Appendix G).

208 6 Ablations

209 The decisive control is — *dynamics*, which sets the integration horizon to zero so the operator is
 210 never applied. The sweep (Fig. 3b, Appendix B) uses a reduced budget at which the full model has
 211 *not* converged (0.233 against 0.247), and undertraining compresses every delta, so the two decisive
 212 variants were re-run at exactly the benchmark budget (3 seeds; Fig. 3c): $T=0$ gives 0.224 ± 0.005
 213 (-0.023) and removing the reaction gives 0.241 ± 0.003 (-0.006). The $T=0$ value falls below *every*
 214 baseline in Fig. 3a, including the non-spatial one.

215 The control is *exactly* parameter-matched: the operator is 9,536 of the model’s 2,198,336 parameters
 216 (0.43%), and $T=0$ leaves them allocated but unused, so architecture, parameter count and optimiser

217 budget are identical and the gap cannot be capacity. That matters because NMO is larger than the
218 graph baselines, so part of its Table 1 margin could be capacity; the $T=0$ control separates those
219 explanations, which is why we weight it above the margin.

220 Two ablations are honestly *null*: removing the biological regularisers changes held-out r by -0.000
221 and removing the PDE constraint terms by -0.000, both within seed noise. At our weights these
222 terms are inert. We report it rather than dropping the rows: a reader is entitled to know that three of
223 the five loss terms in Section 3 do no measurable predictive work, whatever they contribute to the
224 operator’s form. Constraining $\mathbf{D}$ to be isotropic costs -0.004, indicating the anisotropy is used rather
225 than merely available.

226 7 Limitations

227 **One snapshot cannot identify a dynamical system.** Many operators produce the same stationary
228 field, so (1) is identified only up to the quasi-steady-state assumption and the architecture’s inductive
229 bias. Reported diffusion lengths characterise the *fitted operator*, not morphogen transport, and even
230 the Stereo-seq series compares different embryos.

231 **Latent, not molecular; the name describes the operator class.** Latent/pathway correlations
232 show alignment, but our own split-half test rejects the stronger reading (Section 5.5). “Morphogen”
233 names the *operator class*—anisotropic reaction–diffusion—not a validated claim about signalling.
234 Two of three interpretability probes returned negative and a third was underpowered; what survives
235 is a diffusion tensor with a physical length scale and a perturbation half-width consistent with it.

236 **Over-smoothing, worse than the baselines’.** Predicted maps are far more autocorrelated than
237 measured ones (0.936 ± 0.004 vs 0.174): NMO is *worse* than every baseline on Moran’s I by $+0.044$
238 while leading on SSIM by 0.010 . The dissipative operator that delivers the accuracy gain inflates
239 the autocorrelation — one phenomenon, not two. Predicting expression at unmeasured coordinates
240 is supported; delineating sharp boundaries, rare niches or tumour margins is not.

241 **The operator is tissue-specific, which undercuts the motivation.** Discretisation-invariance was
242 the reason to prefer an operator over a graph, yet under transfer the graph baseline wins by $2.8\times$ and
243 $2.0\times$. Until that is fixed—a shared operator across tissues, or $\mathbf{D}_\theta$ conditioned on tissue covariates—
244 this is a within-section model.

245 **Adapted baselines, modest effects, limited scope.** Our *-style* variants implement each method’s
246 architectural core with a common read-out head; fair on *this* task, not a reproduction of published
247 results. The in-domain margin is a few points of Pearson r on one section per condition over few
248 seeds, with no power analysis. Scope is single 2-D sections, ortholog matching by upper-cased
249 symbol, and CPU-scale compute.

250 8 Discussion and Conclusion

251 We asked whether the continuous description developmental biology uses for tissue can be fitted
252 as an operator on spatial transcriptomics. The answer is a qualified yes with a clear boundary. In-
253 domain it helps: the parameter-matched $T=0$ control costs more than the entire margin over the
254 best baseline, and on the one dataset with a real time axis integrating forward beats persistence. The
255 exact spectral solve makes the operator stable and physically readable.

256 Outside that boundary the results are negative, and they are the more useful half of the paper: the
257 operator does not transfer, it is dissipative rather than pattern-forming, its fields are the most over-
258 smoothed of any model tested, and a falsifiable test of the pathway interpretation fails outright. The
259 continuous-operator framing buys in-domain accuracy by specialising to one tissue; portability and
260 mechanism remain unearned, and breaking the identifiability barrier will need dense developmental
261 time series or spatial data paired with targeted perturbation.

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Table 2: Ablations on the mouse-brain Visium section. Each variant removes exactly one ingredient with all else held fixed. Δr is the change in held-out Pearson r relative to the full model. The $- \text{dynamics}$ row is the critical control: it disables the operator entirely ($T=0$) while leaving its parameters allocated, so the two rows are exactly parameter-matched. **This table uses a smaller training budget than Table 1** (fewer epochs and seeds); it is internally consistent, but absolute values are not comparable across the two tables.

variant	params	Pearson $r \uparrow$	Δr
Full model	2.20M	0.233 ± 0.005	–
– dynamics ($T=0$)	2.20M	0.227 ± 0.001	-0.007
– diffusion term	2.20M	0.226 ± 0.004	-0.007
– reaction term	2.19M	0.227 ± 0.001	-0.006
– PDE constraints	2.20M	0.233 ± 0.005	-0.000
– biological regularisers	2.20M	0.233 ± 0.005	-0.000
isotropic D	2.20M	0.229 ± 0.004	-0.004
state-dependent D	2.20M	0.236 ± 0.004	+0.003
discrete GNN operator	0.61M	0.223 ± 0.001	-0.011
latent $C=8$	1.07M	0.213 ± 0.005	-0.021
latent $C=16$	1.30M	0.220 ± 0.002	-0.013
latent $C=64$	5.77M	0.237 ± 0.000	+0.004

Table 3: Processed spatial transcriptomics datasets. Locations and genes are counts after QC and gene selection; extent is the physical bounding box of the section.

dataset	technology	organism	locations	genes	extent (μm)	resolution
merfish_allen_31	MERFISH, 500-gene pane	Mus	91,951	502	9837×6094	single cell
merfish_allen_40	MERFISH, 500-gene pane	Mus	104,461	502	10000×6961	single cell
merfish_allen_50	MERFISH, 500-gene pane	Mus	92,787	501	8806×6604	single cell
mosta_embryo_E10.5	Stereo-seq, DNA nanoba	Mus	18,360	2,099	3500×4950	bin50 (25 μm)
mosta_embryo_E9.5	Stereo-seq, DNA nanoba	Mus	5,898	2,069	1975×2650	bin50 (25 μm)
perturb_norman	Perturb-seq (CRISPRa),	Homo	111,659	2,073	–	single cell
visium_human_breast	Visium (55 μm spots, h	Homo	4,869	2,087	6395×6750	spot (multi-cell)
visium_mouse_brain	Visium (55 μm spots, l	Mus	2,691	2,079	4822×6379	spot (multi-cell)
xenium_mouse_brain	Xenium In Situ, 248-ge	Mus	36,362	248	5212×3128	single cell

A Reproducibility

All datasets download automatically with a recorded SHA256 manifest; every table and figure in this paper is regenerated from run artifacts by a single command; configurations, seeds and checkpoints are released.

Table 4: In-silico morphogen source (*bead implant*). The latent perturbation direction is defined from a random half of each pathway’s genes; the table reports where the *held-out* half ranks in the response magnitude distribution (lower is stronger; 50% is chance). p is a permutation test over random gene sets of matched size. Response half-width is the distance at which the predicted response falls to half its peak. $*p < 0.05$.

pathway	genes	held-out rank (%)	null (%)	p	half-width (μm)
BMP	22	48.6 ± 6.0	49.7	0.475	575 ± 154
FGF	19	72.6 ± 0.8	50.3	0.992	618 ± 58
SHH	15	56.6 ± 6.7	50.4	0.698	466
WNT	26	56.4 ± 3.9	49.9	0.762	649 ± 143

Table 5: Cross-tissue, cross-species transfer: adult mouse brain $\rightarrow$ human breast carcinoma. Shared gene vocabulary: 390 genes. Mean $\pm$ s.d. over seeds.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$
<i>source (in-domain)</i>			
STAGATE-style	0.212 ± 0.001	0.249 ± 0.000	0.358 ± 0.001
NMO (ours)	0.209 ± 0.004	0.251 ± 0.003	0.365 ± 0.004
<i>target (zero-shot)</i>			
STAGATE-style	0.093 ± 0.001	0.397 ± 0.002	0.266 ± 0.002
NMO (ours)	0.033 ± 0.012	0.409 ± 0.005	0.219 ± 0.017
<i>target (decoder-only fine-tune)</i>			
STAGATE-style	0.117 ± 0.007	0.393 ± 0.000	0.365 ± 0.003
NMO (ours)	0.041 ± 0.019	0.406 ± 0.004	0.282 ± 0.017
<i>target (in-domain oracle)</i>			
STAGATE-style	0.147 ± 0.001	0.375 ± 0.000	0.427 ± 0.001
NMO (ours)	0.136 ± 0.001	0.397 ± 0.002	0.329 ± 0.003
<i>target (training-mean floor)</i>			
mean	-0.000 ± 0.000	0.404 ± 0.002	0.254 ± 0.001

Table 6: Cross-resolution transfer: $55 \mu\text{m}$ Visium spots $\rightarrow$ single-cell Xenium. Shared gene vocabulary: 248 genes. Mean $\pm$ s.d. over seeds.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$
<i>source (in-domain)</i>			
NMO (ours)	0.256 ± 0.009	0.246 ± 0.002	0.384 ± 0.007
STAGATE-style	0.242 ± 0.001	0.248 ± 0.001	0.364 ± 0.001
<i>target (zero-shot)</i>			
STAGATE-style	0.282 ± 0.000	1.765 ± 0.001	0.208 ± 0.001
NMO (ours)	0.139 ± 0.015	1.863 ± 0.026	0.125 ± 0.004
<i>target (in-domain oracle)</i>			
NMO (ours)	0.259 ± 0.006	1.752 ± 0.007	0.556 ± 0.006
STAGATE-style	0.256 ± 0.001	1.760 ± 0.002	0.569 ± 0.001
<i>target (training-mean floor)</i>			
mean	0.000 ± 0.000	1.872 ± 0.000	0.101 ± 0.001

Table 7: Agreement between the model’s counterfactual gene–gene responses and measured CRISPRa responses from Norman et al. (2019). **This is an out-of-context probe:** the Perturb-seq data are non-spatial K562 cells while the operator is fitted to human breast tissue, so it tests whether the reaction module encodes generic transcriptional coupling, not whether it recovered tissue-specific signalling. Null is a within-perturbation label permutation; p is a Wilcoxon signed-rank test against that null.

model	#perturb.	Spearman ρ	null	% positive	p
STAGATE-style	7	0.010 ± 0.000	+0.004	64%	0.633
NMO (ours)	7	-0.018 ± 0.009	-0.009	57%	0.844

Table 8: Developmental forecasting on Stereo-seq: the E9.5 field is encoded, integrated forward for T operator steps, and scored against the measured E10.5 field. **Consecutive stages are different embryos**, so there is no cell-to-cell correspondence and the comparison is made at the level of the rasterised field after isotropic normalisation to a common frame; it inherits the error of that coarse registration. *Persistence* predicts $E10.5 = E9.5$ and is the reference the operator must beat for its dynamics to carry temporal information.

model	horizon	field $r \uparrow$	field RMSE $\downarrow$	genes with $r > 0$
Persistence (no dynamics)	–	0.064	0.346	61%
Spatial-mean predictor	–	0.000	0.257	4%
NMO	$T=0$	0.082 ± 0.001	0.320	74%
NMO	$T=2$	0.089 ± 0.000	0.305	72%
NMO	$T=4$	0.092 ± 0.000	0.297	73%
NMO	$T=8$	0.094 ± 0.000	0.289	72%
NMO	$T=16$	0.095 ± 0.001	0.283	72%
NMO	$T=32$	0.090 ± 0.003	0.285	70%

360 B Ablation table

361 C Dataset inventory

362 D In-silico morphogen source: per-pathway results

363 E Cross-tissue transfer

364 F Cross-resolution transfer

365 G Perturb-seq consistency of the reaction Jacobian

366 H Developmental forecasting: full results

367 Every experiment so far fits a *stationary* operator, because the data are snapshots. The Stereo-seq
368 developmental series is the single dataset here with a real time axis, so it is the only place we can ask
369 the operator to act as a forward-time predictor. We encode the E9.5 field, integrate for T operator
370 steps, and score the decoded prediction against the measured E10.5 field (Table 8).

371 The comparison must be read carefully. **Consecutive MOSTA stages are different embryos**, not
372 the same embryo imaged twice, so no cell-to-cell correspondence exists and none can be manu-
373 factured by registration. We therefore evaluate at the level of the rasterised field after isotropic
374 normalisation to a common frame, and the score inherits the error of that coarse alignment. The
375 reference to beat is *persistence* — predicting $E10.5 = E9.5$. If the learned dynamics carried no
376 temporal information, NMO could not exceed it.

Persistence attains field-level $r = 0.064$, which is itself low and quantifies how much the embryo changes (and how crude the registration is) between these stages. Sweeping the horizon rather than tuning it, NMO reaches 0.095 at $T = 16$. We report the full sweep so the horizon is visible as a reported quantity rather than a hyperparameter chosen against the test field. This experiment is the weakest in the paper in terms of statistical power, and we present it as an existence check on forward-time behaviour rather than as a forecasting benchmark.

I Counterfactual perturbation: full protocol and results

We perform the computational analogue of a morphogen bead implant: inject the latent direction that maximally raises a pathway’s decoded activity into a small disc, integrate the operator, and read out the spatial response. The design is deliberately falsifiable and split-half—the direction is defined from a random half of each pathway’s genes and enrichment is scored on the *held-out* half—with a permutation null over matched random gene sets.

This test fails, and that is the primary outcome of the experiment. Across 4 pathways and 2 seeds, 0 reach significance; the smallest per-pathway p is 0.47 and the mean held-out rank is 59% against a 50% chance level, so if anything held-out pathway members respond *less* than random genes (Appendix D).

The result rules out a reading the correlations in Fig. 7b would otherwise invite. Moving the latent direction that most raises one half of a pathway does not preferentially move the other half, so those correlations are consistent with the field encoding coarse anatomy that pathway scores also track, rather than a pathway-specific representation. We therefore do not claim NMO recovers signalling programmes. The experiment does establish a scale: the response decays to half its peak within $577\ \mu\text{m}$, matching the diffusion lengths read off $\mathbf{D}_\theta$. It is the *specificity*, not the range, that the test rejects.

Separately, we ask whether the reaction module encodes real transcriptional coupling by comparing the model’s counterfactual gene–gene responses against measured CRISPRa responses [Norman et al., 2019]. **This is an out-of-context probe and we treat it as such:** Norman et al. profiled non-spatial K562 cells, whereas our operator is fitted to tissue. We fit on the human breast section so at least species and a proliferative context are shared, report a permutation null, and report the same quantity for every baseline.

This probe is underpowered and we treat it as inconclusive. Only 7 of Norman’s single-gene perturbations fall inside the spatial model’s gene vocabulary (339 genes are shared overall): HVG selection on a breast-cancer section and a CRISPRa screen in K562 agree on very few targets. With $n = 7$ the test cannot separate absence of effect from absence of power. The observed value ($\rho = -0.018 \pm 0.009$ against a null of -0.009 , Wilcoxon $p = 0.84$) is consistent with zero and we draw no conclusion from it in either direction. We report it because the experiment was pre-specified, not because it decided anything; a properly powered version would need a screen whose targets were chosen to overlap the spatial panel.

J Ablation figure

K Benchmark summary figure

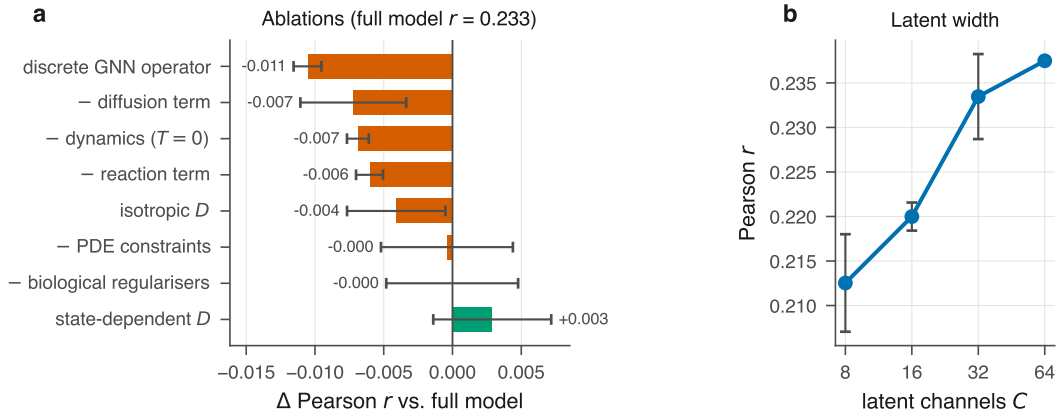


Figure 8: **Ablations.** (a) Change in held-out Pearson r when each ingredient is removed. (b) Latent-width sweep.

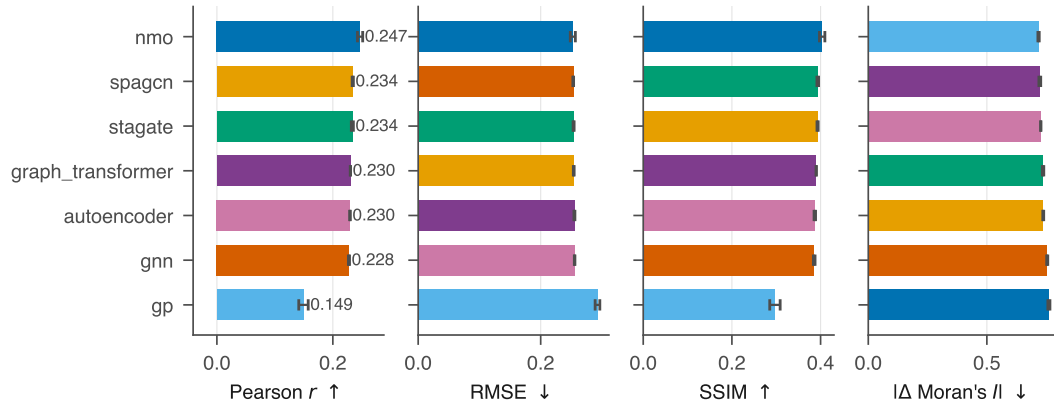


Figure 9: Benchmark of Table 1 across four complementary metrics. Moran's I error (right) is the only axis on which NMO is not best.